

Heart Failure

1. Definition

Heart failure (HF) is a clinical syndrome in which the heart is unable to pump sufficient blood to meet the metabolic needs of the body.

2. Classification

A. By Time Course

- Acute Heart Failure: Sudden onset (e.g., due to MI, acute valve failure).
- Chronic Heart Failure: Long-standing, progressive.

B. By Side of Heart Involved

- Left-sided HF: Pulmonary congestion predominates.
- Right-sided HF: Systemic venous congestion predominates.

C. By Ejection Fraction

- HFrEF (EF $\leq$ 40%): Systolic dysfunction (impaired contraction)
- HFpEF (EF $\geq$ 50%): Diastolic dysfunction (impaired relaxation)
- HFmrEF (EF 41–49%): Mildly reduced EF

3. Etiology

Common Causes:

- Ischemic heart disease (most common)
- Hypertension
- Valvular heart disease
- Cardiomyopathies (dilated, hypertrophic, restrictive)
- Arrhythmias (AF, tachycardia-induced)
- Congenital heart disease

4. Pathophysiology

1. Decreased cardiac output → reduced tissue perfusion
2. Neurohormonal activation:
 - ↑ Sympathetic activity → vasoconstriction, tachycardia
 - ↑ RAAS → sodium & water retention
 - ↑ ADH → further fluid retention
3. Myocardial remodeling → hypertrophy, dilation, worsening HF

5. Clinical Features

A. Left-Sided Failure:

- Dyspnea (especially on exertion)
- Orthopnea
- Paroxysmal nocturnal dyspnea

- Fatigue, weakness
- Cough, frothy sputum
- Basal lung crepitations

B. Right-Sided Failure:

- Peripheral edema
- Hepatomegaly
- Ascites
- Jugular venous distension (JVD)
- Weight gain

6. Investigations

ECG: LVH, ischemia, arrhythmia

Chest X-ray: Cardiomegaly, pulmonary congestion

Echocardiography: EF measurement, structural abnormalities

BNP / NT-proBNP: Elevated in HF

Blood tests: Renal function, electrolytes, thyroid function

Cardiac MRI: Myocardial fibrosis, infiltration (if indicated)

7. Management

A. General / Lifestyle:

- Sodium restriction (<2 g/day)
- Fluid restriction (in severe cases)
- Weight monitoring
- Stop smoking, alcohol
- Exercise as tolerated

B. Pharmacological:

ACE inhibitors / ARBs (Enalapril, Losartan): ↓ Afterload, prevent remodeling

Beta-blockers (Carvedilol, Metoprolol): ↓ Sympathetic drive

Diuretics (Furosemide): Relieve congestion

Aldosterone antagonists (Spironolactone): ↓ Mortality

SGLT2 inhibitors (Dapagliflozin, Empagliflozin): Improve outcomes

Digoxin (in select cases): ↑ Contractility

8. Complications

- Arrhythmias (AF, VT)
- Thromboembolism
- Renal failure
- Cardiogenic shock
- Death (if untreated or advanced)

9. Prognosis

Poor if untreated — 5-year mortality ~50%

Improved survival with guideline-directed therapy

10. Nursing Considerations

- Monitor daily weight and I/O
- Check for edema, lung sounds
- Administer medications as prescribed
- Educate patient on diet, adherence, and signs of worsening HF